The Nature of Econometrics and the Application of Econometric Models in Economics and Business

These notes accompany Lecture 1. They combine the slide material with the explanations given in class, so you can follow the argument without the slides in front of you. Terms printed in bold are defined in the glossary at the end.

Learning outcomes – after this lecture you should be able to:

1. Explain what econometrics is, and how it differs from economic theory, mathematical statistics, and machine learning.
2. Explain why converting qualitative statements into quantitative ones is the central task of the discipline.
3. Describe the seven steps of an applied econometric analysis and say what can go wrong at each.
4. Distinguish correlation from causation, and explain the role of the ceteris paribus condition.
5. Identify the four structures of economic data and recognise which one a given dataset has.

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1. What is econometrics?

1.1 Why economic theory alone is not enough

Economists build models to explain relationships that recur consistently. An economic model links one or more economic variables to other economic variables. For example, economists connect the amount individuals spend on consumer goods to their disposable income and wealth, and expect consumption to rise as either increases – that is, the relationship is positive.

The difficulty is that there are often several competing models explaining the same recurring relationship, and few, if any, of them tell us anything about the size of the association. Theory typically delivers a sign, not a number.

But a sign is rarely what a decision-maker needs. What matters most in practice is not the explanation of the relationship but the magnitude of the association between the variables. A central bank deciding on interest rates does not merely want to know that raising rates lowers inflation; it needs to know by how much, and after how long. Without a number, the theory cannot be acted upon.

> **The motivating question for this course**
>
> What impact on price growth can be expected from the central bank's interest rate hike?
>
> Theory answers "inflation should fall". That is not enough to set policy. Econometrics exists to supply the missing magnitude.

1.2 From qualitative to quantitative statements

A key objective of econometrics is to convert qualitative statements into quantitative ones. The difference is easiest to see side by side:

Qualitative statement	Quantitative statement
Higher interest rates reduce inflation	Inflation decreases by 0.3 p.p. a year after a 1 p.p. increase in the interest rate
The relationship between taxes and GDP is negative	GDP decreases by 0.3% for every 1 p.p. increase in the VAT rate

Both columns say something true and both come from economic reasoning. Only the right-hand column can be used to make a decision, to be tested against evidence, or to be proved wrong.

How is the conversion done?

By combining two ingredients: statistical data, and statistical and mathematical techniques applied to that data. This is where the three parent disciplines meet:

- **Economics** – supplies theory and questions: what should happen, why might it happen, which mechanisms matter?
- **Statistics** – supplies evidence: what actually happened, how much variation is there, how noisy is the evidence?
- **Mathematics** – supplies estimation: how do we quantify the relationship?

Econometrics sits where all three overlap. It is the discipline of *learning* from data: what relationship is in the data, how large is it, and how certain are we?

1.3 A definition

Literally, "econometrics" means measurement in economics (from the Greek metrics, measurement). More usefully:

> **Definition**
>
> Econometrics combines economic theory with mathematical and economic statistics in order to:
>
> - analyse economic relationships,
> - test economic theories,
> - evaluate government or business policies.

The study of the methods that allow us to quantify economic relationships using actual data is what we call econometrics. Its purpose is to turn theoretical economic models into usable tools for policymaking and decision-making – to attempt to prove or disprove economic propositions using evidence.

The most visible application is forecasting macroeconomic variables such as interest rates, inflation and GDP. But econometric methods are used far beyond macroeconomic forecasting: to study the effect of political campaign expenditure on voting outcomes, the effect of school spending on student performance, the drivers of business performance, and much else.

Ragnar Frisch and the founding of the discipline

The word and the discipline are associated with Ragnar Frisch, who received the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel in 1969, jointly with Jan Tinbergen – the first year the prize was awarded.

Writing in the first issue of Econometrica in 1933, Frisch made the point that still defines the field: econometrics is not the same as economic statistics, nor is it identical with general economic theory, nor is it simply the application of mathematics to economics. Each of the three viewpoints – statistics, economic theory, and mathematics – is necessary but not by itself sufficient for a real understanding of quantitative relations in modern economic life. It is the unification of all three that is powerful, and it is that unification that constitutes econometrics.

Source: Frisch, R. (1933). Editor's Note. Econometrica, 1(1), 1–4.

1.4 Econometrics is not machine learning

Econometrics is not merely statistical inference, nor is it the use of big data to quantify empirical relationships. The reason is fundamental: data do not speak for themselves.

A relationship found in the data does not tell us why it exists, and the same data can be consistent with several competing explanations. Economic reasoning is needed to interpret statistical evidence.

The distinction in one line

Big data tells you what will happen. Econometrics tells you what would happen if you changed something.

A model can predict inflation beautifully and still tell you nothing about what the central bank should do – because the moment the bank acts, the correlations the model learned may no longer hold. That is the Lucas critique, and no amount of data volume fixes it.

1.5 Why not just mathematical statistics?

Econometrics differs from mathematical statistics because of the non-experimental nature of economic data.

Controlled experiments, in which a single factor is varied and everything else is held fixed, are the norm in the natural sciences but are often impossible, prohibitively expensive, or ethically unacceptable in economics. Non-experimental data – also called observational data – are not generated by controlled experiments on individuals, firms, or segments of the economy; they are simply what the economy happened to produce.

Experimental data are routinely collected in laboratories in the natural sciences, but are far harder to obtain in the social sciences. Some social experiments can be devised, but conducting the kind of controlled experiment that would settle most economic questions is often impossible, prohibitively expensive, or ethically unacceptable.

Econometricians have therefore borrowed from mathematical statisticians wherever possible, and adapted the tools to the data they actually have. There is also a strong recent trend towards estimating models built around natural experiments – situations in which circumstances happen to produce something close to random variation.

1.6 Where do we see econometrics?

Almost everywhere a decision depends on the size of an effect:

Field	Typical question
Central Bank	Will raising the policy rate reduce inflation?
Government	Does a higher minimum wage reduce poverty?
Education	Does more education increase earnings?
Health	Does a new treatment improve outcomes?
Business	Will lowering the price increase sales?
Macroeconomics	What will happen to GDP next year?

1.7 Correlation versus causation – the fundamental issue

A warning example: shoe size and reading ability

Primary-school teachers have long observed that their pupils' reading ability tends to increase with their shoe size. Should policymakers therefore fund research into making children's feet bigger?

Obviously not. Older children have bigger feet, but they also have more developed brains. The natural development of children explains why shoe size and reading ability increase together – that is, why they are positively correlated. But bigger feet do not cause better reading.

In economics, correlations are everywhere. Determining whether a correlation between two variables reflects a causal relationship is rarely as easy as it is in the shoe-size example.

A problem economists actually face

Economic theory says that raising the policy interest rate reduces inflation. But consider when a central bank actually raises rates. It raises them precisely when it detects significant price pressure – from very strong domestic demand, say, or from rising global commodity prices – and therefore expects inflation to increase in the near future.

The rate rise then affects the economy not immediately but with a time lag. So what do we observe in the data? Inflation rises after the rate hike, exactly as the bank had feared. Does this mean higher interest rates cause higher inflation?

No. That is only a correlation in the data. Without the rate hike, inflation would have risen even more. The true causal effect of the interest rate on inflation is negative, even though the raw correlation is positive.

> **Why this matters**
>
> The sign of the correlation in the data and the sign of the causal effect can be opposite.
>
> Distinguishing one from the other is the central problem econometrics exists to solve.

Why is economic data so difficult?

When we observe inflation rising, the cause might be strong demand that production cannot meet; it might be policy, such as an increase in indirect taxes; it might be shocks, such as a commodity price spike or a logistical disruption; or it might be something else entirely.

Economic outcomes are generated by many forces at once – economic behaviour, policy, shocks, institutions – and are then filtered through imperfect measurement before we ever see them as data on GDP, inflation, wages, prices or employment. This is precisely why isolating the causal relationship between an outcome variable (inflation) and a variable of interest (the policy rate) is so hard.

1.8 Ceteris paribus

The economist's goal is often to infer the causal effect of one variable (such as education) on another (such as worker productivity). An association between two or more variables may suggest, but does not establish, a causal effect – correlation does not imply causality.

The notion of ***ceteris paribus*** – "other relevant factors being equal", or "all else equal" – plays the key role in causal analysis. Most economic questions are ceteris paribus by nature.

In demand analysis, for example, we want the effect of a change in a good's price on the quantity demanded, holding income, the prices of other goods, and individual tastes fixed. If those other factors are not held fixed, we cannot isolate the causal effect of the price change on quantity demanded. Holding other factors fixed is equally critical for policy analysis.

Much of the technical machinery in the rest of this course exists for one purpose: to hold other factors fixed when we cannot do so experimentally.

2. Steps in empirical economic analysis

Economists use empirical analysis to test hypotheses and to quantify economic relationships. Applied econometric work has – or at least should have – a model from economic theory as its starting point. From that theory, the task is to formulate an econometric model in an empirically testable form, collect suitable data, estimate the model, and then check it.

The process is a loop, not a straight line. If the specification tests and diagnostics indicate that the model is not appropriate, the econometrician returns to the model formulation stage, revises the model, and repeats the procedure. Only when the model passes its diagnostics can hypotheses be tested and the model used for prediction and policy advice.

> **The cycle at a glance**
>
> Economic theory → Econometric model → (Data) → Estimation → Model diagnostics
>
> If the model is adequate → test hypotheses → use the model for prediction and policy.
>
> If the model is not adequate → return to the econometric model and revise.

Step 1. Formulation of the question of interest

The first step in any empirical analysis is the careful formulation of the question. The question may test an aspect of economic theory, or it may concern the effects of a policy. In principle, econometric methods can address a very wide range of questions:

- Does an increase in government spending have an impact on economic growth?
- What is the impact of a policy interest rate change on inflation?
- Do less capitalised banks react differently to a monetary policy shock?
- Does job training have an impact on worker productivity?
- Does an additional year of schooling have an impact on one's wage?
- Is there an impact of legalised abortion on crime?
- Does a price increase have an impact on a firm's revenues?

Step 2. Specification of the economic model

(a) Sometimes a formal economic model is constructed

In some cases – particularly when testing economic theories – a formal economic model is built. An economic model consists of mathematical equations describing the relationships involved.

Demand theory, for instance, assumes individuals allocate their budget across a set of goods so as to maximise utility for a given set of prices. Utility maximisation subject to the budget constraint implies a system of demand equations relating the quantity consumed of each good to the consumer's budget, to prices, and to individual characteristics that affect taste:

$$q_1 = f_1(x, p_1, \ldots, p_k, age, educ)$$

$$\vdots$$

$$q_k = f_k(x, p_1, \ldots, p_k, age, educ)$$

(b) More often, theory is used informally – or we rely on intuition

Formal modelling is sometimes the starting point, but it is more common to use economic theory less formally, or to rely on intuition entirely.

Suppose a labour economist wants to examine the effect of job training on worker productivity. Little formal theory is needed here. Basic economic understanding is enough to recognise that factors such as education, experience and training affect worker productivity; and economists are well aware that workers are paid broadly in line with their productivity. That reasoning alone leads to a model:

$$wage = f(educ, exper, training)$$

Step 3. Formulation of the econometric model

Once the economic model is specified, it must be turned into an econometric model. Two decisions are unavoidable.

- **The functional form** of f(·) has to be specified before any estimation can be done. The wage equation, for example, may be given a linear form.
- **Unobserved variables** must be dealt with. Some factors – the quality of a person's education, their innate ability, their family background – cannot reasonably be observed, and many others we cannot even list. They must nevertheless be accounted for somehow.

A complete econometric model might then look like this:

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 training + u$$

where *educ* and *exper* are years of education and experience, and *training* indicates whether the person had additional training.

- $\beta_0, \beta_1, \beta_2$ *and* β_3 are the **parameters** of the model. They describe the direction and strength of the relationship between the wage and each factor.
- u is the **error term** (or **disturbance term**). It contains innate ability, quality of education, family background, and every other unobserved factor influencing the wage.

Dealing with the error term is perhaps the single most important component of any econometric analysis – a theme that will recur throughout this course.

Step 4. Data collection

An empirical analysis, by definition, requires data. Data on the relevant variables have to be collected, and the usual sources fall into three groups:

- **Microeconomic data** – typically surveys: individuals' consumption of goods, prices, budgets and characteristics; workers' wages, education, experience and training; business performance indicators; and so on.
- **Macroeconomic data** – statistical offices, central banks, the IMF, the World Bank, the OECD, CEIC and similar providers.
- **Financial data** – commercial banks, Bloomberg, Reuters, stock exchanges.

Step 5. Estimation and diagnostics

Once data have been collected, econometric methods are used to estimate the parameters of the econometric model and to test hypotheses of interest formally. Ordinary least squares (OLS) is the workhorse we begin with.

Estimation is not the end. The applied econometrician must then run specification tests to confirm that the model used was the appropriate one, together with diagnostic checks on the performance and accuracy of the estimation procedure. This is the stage at which the loop may send you back to Step 3.

Step 6. Test hypotheses and interpret the model

If the diagnostics suggest the model is adequate, the next step is hypothesis testing: does the evidence support the theoretical prediction? In the wage equation we might hypothesise that training increases a worker's wage. That hypothesis is equivalent to a statement about a single parameter:

$$H: \beta_3 > 0$$

Interpretation then matters as much as the test. Always check that the estimated parameters make sense. What does the magnitude of the estimate actually mean in economic terms? Is it plausible? It is not sufficient to conclude that an effect is statistically significant – an effect can be statistically significant and economically trivial.

Step 7. Present the results

Finally, the results must be communicated – as a journal article, a research paper, a policy report, or an analytical note. The format differs, but the obligation is the same: the reader must be able to judge whether your conclusion is warranted.

3. The structure of economic data

Economic datasets come in several forms. Some econometric methods apply straightforwardly across all of them, but the special features of each structure matter enormously, and using a method designed for one structure on another is a common and serious error. There are four structures to know.

3.1 Cross-sectional data

A cross-sectional dataset consists of a sample of individuals, households, firms, cities, states, countries, or other units, taken at a given point in time.

Cross-sectional variables are denoted by the subscript i, with i running from 1 to N, where N is the number of cross-sections. If Y denotes income collected for N individuals, we write:

$$Y_i \quad \text{for } i = 1, 2, 3, \ldots, N$$

Two features matter. First, we can often assume the data were obtained by random sampling from the underlying population (for survey data in particular). Second, the ordering of the data does not matter for the analysis – it makes no difference which person is labelled observation 1 and which observation 2. That property follows directly from random sampling.

In economics, cross-sectional analysis is mainly associated with applied microeconomics: labour economics, state and local public finance, business economics, demographic economics and health economics.

Table 1. Wages and other individual characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
⋮	⋮	⋮	⋮	⋮	⋮
526	3.50	14	5	1	0

wage is in dollars per hour; educ is years of education; exper is years of potential labour force experience. female and married are binary (zero–one) variables recording qualitative features of the individual.

Table 2. GDP per capita, life expectancy and spending on research, 2021

obsno	country	gdp_ppp	life_exp	RandD_share
1	Belarus	27 611.18	72.07	0.46
2	Lithuania	45 873.53	74.04	1.10
⋮	⋮	⋮	⋮	⋮
120	Germany	65 530.77	80.79	3.13

3.2 Time series data

Time series data track variables over time – stock prices, the money supply, the consumer price index, GDP. They are denoted with the subscript *t*. If *Y* denotes a country's GDP from 1990 to 2025, we write:

$$Y_t \quad for\ t = 1, 2, 3, \ldots, T$$

where $t = 1$ corresponds to 1990 and $t = T = 36$ to 2025.

Three features distinguish time series from cross-sections:

- Observations can rarely be assumed independent across time. What happened last quarter tells you a great deal about this quarter.
- Ordering is essential – unlike a cross-section, you cannot shuffle the rows.
- Data frequency varies: daily, weekly, monthly, quarterly, annual. Many weekly, monthly and quarterly series display a strong seasonal pattern, which can be an important factor in the analysis.

Time series data are mainly associated with macroeconomic applications.

Table 3. US macroeconomic data

period	rgdp	cpi	fedrate	eurusd	oil_price
2000Q1	13 878.15	100.00	5.68	0.99	26.85
2000Q2	14 130.91	100.70	6.27	0.93	26.52
2000Q3	14 145.31	101.52	6.52	0.91	30.20
⋮	⋮	⋮	⋮	⋮	⋮
2026Q2	24 270.60	194.48	3.63	1.17	104.43

3.3 Pooled cross sections

Some datasets have both cross-sectional and time series features. Suppose two household surveys are taken, one in 2015 and one in 2025. In 2015 a random sample of households is surveyed for income, savings, family size and so on; in 2025 a new random sample is taken using the same questions. Combining the two years gives a pooled cross section, which increases the sample size.

> **The key point**
>
> In a pooled cross section, the households observed in the two years are different. We are not following the same households over time.

Pooling cross sections from different years is often an effective way of analysing the effects of a new government policy: collect data from the years before and after a key policy change, then examine how the relationship of interest has shifted.

Table 4 shows house prices from 1993 and 1995, before and after a reduction in property taxes in 1994. Observations 1–250 are houses sold in 1993; observations 251–520 are the 270 houses sold in 1995. Although the storage order is not crucial, keeping track of the year of each observation is essential – which is why year is entered as a separate variable.

Table 4. Two years of housing prices

obsno	year	hprice	proptax	sqrft	bdrms	bthrms
1	1993	85 500	42	1 600	3	2.0
2	1993	67 300	36	1 440	3	2.5
⋮	⋮	⋮	⋮	⋮	⋮	⋮
251	1995	65 000	16	1 250	2	1.0
252	1995	182 400	20	2 200	4	2.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
520	1995	57 200	16	1 100	2	1.5

3.4 Panel data

A panel dataset consists of a time series for each cross-sectional member of the dataset. Because panel data have both dimensions, they carry both subscripts, *i* and *t*:

$$Y_{it} \quad \text{for } t = 1, \ldots, T \text{ and } i = 1, \ldots, N$$

Examples: wage, education and employment history for a set of individuals followed over ten years; investment and financial data for the same firms over five years; GDP and money supply for 20 countries over 20 years.

> **Panel data versus pooled cross sections**
>
> The key distinction is that in a panel the same cross-sectional units – the same individuals, firms or countries – are followed over the time. In a pooled cross section they are different units each period.

Because panel data require the same units to be observed repeatedly, they are harder to obtain than pooled cross sections. The payoff is substantial: having multiple observations on the same unit allows us to control for certain unobserved characteristics of that unit – which, as Section 1.8 explained, is exactly the problem econometrics is trying to solve. Specific econometric techniques exist to exploit this, and we return to them later in the course.

Table 5. Data on commercial banks

date	bank	Interbank_rate	lending_rate	npl	liquidity	assets
2013m1	108	35.3	37.1	4.3	52.1	451 458
2013m1	110	35.3	43.2	1.6	76.9	47 135
⋮	⋮	⋮	⋮	⋮	⋮	⋮
2015m1	108	45.2	51.9	12.4	224.1	760 364
2015m1	110	45.2	50.9	4.0	57.1	81 192
⋮	⋮	⋮	⋮	⋮	⋮	⋮
2017m12	108	9.7	10.6	21.8	74.2	757 523
2017m12	110	9.7	12.9	4.5	131.2	109 398

Note that the interbank rate is identical for both banks within each month – it is a market-wide rate – while the lending rate, non-performing loans, liquidity and assets vary across banks. The same banks are observed repeatedly, which is what makes this a panel rather than a pooled cross section.

3.5 Summary of the four data structures

Structure	Subscript	Units observed	Does order matter?
Cross-section	i	Many units, one point in time	No
Time series	t	One unit, many periods	Yes
Pooled cross section	i, t	Different units each period	Year must be recorded
Panel	i, t	Same units, followed over time	Yes, within each unit

4. Summary

4.1 What do econometricians actually do?

Everything in this course falls into one of three activities.

1. Describe. What does the data look like? What relationship do we see, how strong is it, how much variation is there? Most empirical work begins here – but notice that description makes no claim about *why*.

2. Identify. What can we say about the underlying economic mechanism – can we interpret an effect causally? This is the hard part, and it is where econometrics separates itself from data analysis. The question is not answered by more data or a better fit; it is answered by an argument about where the variation came from, and whether anything else could explain what we see.

3. Predict. What is likely to happen next, how uncertain is the forecast, and what scenarios are plausible? Note that prediction does not require identification – you can forecast well without understanding the mechanism. But if you want to know what happens *when policy changes*, you need the middle box first.

4.2 Economics, statistics and econometrics

Each discipline asks a different question about the same pair of variables:

Discipline	The question it asks
Economics	Why might the interest rate affect inflation?
Statistics	Are the interest rate and inflation associated in the data?
Econometrics	What can we learn about the economic relationship from the data? Does the interest rate actually cause lower inflation?

Economics supplies the theory – a reason to expect a relationship at all, and in which direction. Without it, you are fishing. Statistics tells you whether the pattern is really in the data or could plausibly be chance; that is a question about the sample, not about the world. Econometrics is where the two meet, and it asks the question neither can answer alone: given a theory about a mechanism and a body of noisy data, what can we actually learn – and can we defend calling it causal?

The lecture in three sentences

Economics without data is untested.

Statistics without theory is pattern-matching.

Econometrics is the discipline of making the two speak to each other.

Key terms

Term	Meaning
Econometrics	The study of the methods that enable us to quantify economic relationships using actual data. Econometrics combines economic theory, mathematical and economic statistics to analyse economic relationships, test economic theories, and evaluate government or business policies.
Economic model	A set of mathematical equations linking one or more economic variables to others.
Econometric model	An economic model given a specific functional form and an error term, so that its parameters can be estimated from data.
Parameters	The coefficients (β_0, β_1, …) describing the direction and strength of each relationship in the model.
Error term (disturbance term)	The term u, containing all unobserved factors that influence the dependent variable.
Ceteris paribus	"Other relevant factors being equal." The condition under which a causal effect is defined.
Correlation	A tendency for two variables to move together. It does not imply causation.
Causal effect	The change in an outcome that results from changing one variable while holding all other factors fixed.
Non-experimental (observational) data	Data not generated by a controlled experiment – the normal case in economics.
Natural experiment	A situation in which circumstances produce variation resembling random assignment.
Lucas critique	The argument that relationships estimated from past data may break down once policy changes, because agents change their behaviour.
Cross-sectional data	Many units observed at a single point in time; ordering is irrelevant.
Time series data	One unit observed over many periods; ordering is essential and observations are rarely independent.
Pooled cross section	Cross-sections from different periods combined, with different units observed each period.
Panel (longitudinal) data	The same units followed over several periods, carrying both i and t subscripts.

Review questions

Use these to check your understanding before the seminar.

1. Why is it not enough for economic theory to tell us the sign of a relationship? Give an example of a decision that requires a magnitude.
2. Convert the following qualitative statement into a quantitative one, inventing plausible numbers: "A weaker exchange rate raises inflation."
3. Frisch argued that econometrics is not the same as economic statistics, economic theory, or mathematics. Explain in your own words what he thought each one is missing on its own.
4. A colleague says: "With enough data, machine learning will make econometrics unnecessary." Give the strongest counter-argument you can, using the Lucas critique.
5. Explain why the raw correlation between the policy interest rate and inflation can be positive even though the causal effect is negative.
6. Why can econometricians rarely run controlled experiments? Give one economic question where an experiment would be impossible, one where it would be too expensive, and one where it would be unethical.
7. In the wage equation $wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 training + u$, name three factors contained in u. Why is the treatment of u the most important part of the analysis?
8. State the hypothesis "training raises wages" as a statement about a parameter. What would it mean for this effect to be statistically significant but economically unimportant?
9. For each dataset, identify the structure (cross-section, time series, pooled cross section, or panel): (a) monthly Latvian CPI, 2005–2025; (b) a 2024 survey of 800 firms; (c) surveys of 500 different households in 2015 and 600 different households in 2020; (d) annual balance sheets of the same 30 banks, 2015–2024.
10. Why does the ordering of observations matter for time series but not for a cross-section obtained by random sampling?
11. What is the single practical advantage of panel data over a pooled cross section, and why does it matter for causal analysis?

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapter 1 covers the material of this lecture.

Asteriou, D., Hall, S.G. Applied Econometrics – Preface and Chapter 2.